

# Homework 2

## Linear Statistical Models

Mohammad Shaan  
BS2419

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1. Consider the linear model

$$Y = X\beta + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I_n),$$

where  $X$  is  $n \times p$  with  $\text{rank}(X) \leq p$ . Define

$$\mu = E(Y), \quad \theta = X^T \mu.$$

(a) Show that the least squares estimator of  $\mu$  under model (1) is

$$\hat{\mu} = P_X Y,$$

where  $P_X$  denotes the orthogonal projection onto the column space of  $X$ .

(b) Show that  $\theta$  is estimable and its BLUE is given by

$$\hat{\theta} = X^T Y.$$

(c) Show that if  $a^T \beta$  is estimable, then the BLUE of  $a^T \beta$  is  $w^T \hat{\theta}$  for some  $w \in \mathbb{R}^p$ .

### Solution:

(a) Since

$$\mu = E(Y) = X\beta,$$

we have

$$\mu \in \mathcal{C}(X),$$

where  $\mathcal{C}(X)$  denotes the column space of  $X$ .

The least squares estimator of  $\mu$  is a solution to

$$\min_{m \in \mathcal{C}(X)} \|Y - m\|^2.$$

Let  $P_X$  denote the orthogonal projection matrix onto  $\mathcal{C}(X)$ . Then

$$Y = P_X Y + (I - P_X)Y,$$

where

$$P_X Y \in \mathcal{C}(X) \quad \text{and} \quad (I - P_X)Y \in \mathcal{C}(X)^\perp.$$

For any  $m \in \mathcal{C}(X)$ , since

$$P_X m = m \quad \text{and} \quad (I - P_X)m = 0,$$

we have

$$\begin{aligned} Y - m &= P_X Y + (I - P_X)Y - m \\ &= (P_X Y - m) + (I - P_X)Y. \end{aligned}$$

The two terms on the right-hand side are orthogonal. Hence,

$$\|Y - m\|^2 = \|P_X Y - m\|^2 + \|(I - P_X)Y\|^2.$$

The second term does not depend on  $m$ . Therefore,

$$\begin{aligned} \arg \min_{m \in \mathcal{C}(X)} \|Y - m\|^2 &= \arg \min_{m \in \mathcal{C}(X)} \|P_X Y - m\|^2 \\ &= P_X Y. \end{aligned}$$

Thus the least squares estimator of  $\mu$  is

$$\boxed{\hat{\mu} = P_X Y}.$$

(b) Recall that

$$\theta = X^T \mu = X^T X \beta.$$

Consider the estimator

$$\hat{\theta} = X^T Y.$$

Then

$$E(\hat{\theta}) = E(X^T Y) = X^T E(Y) = X^T X \beta = \theta.$$

Thus  $\hat{\theta} = X^T Y$  is a linear unbiased estimator of  $\theta$ , and hence  $\theta$  is estimable.

We now show that  $\hat{\theta}$  is the BLUE.

Let  $AY$ , where  $A$  is a  $p \times n$  matrix, be any other linear unbiased estimator of  $\theta$ . Then

$$E(AY) = AX\beta = X^T X \beta \quad \text{for all } \beta,$$

which implies

$$AX = X^T X.$$

Let

$$Q = A - X^T.$$

Then

$$QX = AX - X^T X = 0.$$

Now,

$$\text{Var}(AY) = \sigma^2 AA^T,$$

while

$$\text{Var}(X^T Y) = \sigma^2 X^T X.$$

Since

$$A = X^T + Q,$$

we obtain

$$\begin{aligned} \text{Var}(AY) &= \sigma^2 (X^T + Q)(X + Q^T) \\ &= \sigma^2 (X^T X + X^T Q^T + QX + QQ^T). \end{aligned}$$

Since  $QX = 0$ , we also have

$$X^T Q^T = (QX)^T = 0.$$

Therefore,

$$\text{Var}(AY) = \sigma^2 X^T X + \sigma^2 QQ^T.$$

Hence

$$\boxed{\text{Var}(AY) - \text{Var}(X^T Y) = \sigma^2 QQ^T \succeq 0.}$$

Equivalently, for every  $j = 1, \dots, p$ ,

$$e_j^T [\text{Var}(AY) - \text{Var}(X^T Y)] e_j = \sigma^2 e_j^T QQ^T e_j \geq 0.$$

Thus every component of  $X^T Y$  has variance no larger than the corresponding component of any other linear unbiased estimator of  $\theta$ . Therefore,

$$\boxed{\hat{\theta} = X^T Y}$$

is the BLUE of  $\theta$ .

(c) We first recall that  $a^T \beta$  is estimable if and only if

$$a \in \mathcal{C}(X^T).$$

Thus, since  $a^T \beta$  is assumed to be estimable, there exists  $u \in \mathbb{R}^n$  such that

$$a = X^T u.$$

Consequently,

$$a^T \beta = u^T X \beta = u^T \mu.$$

Consider the estimator

$$u^T \hat{\theta} = u^T X^T Y.$$

Then

$$\begin{aligned} E(u^T \hat{\theta}) &= u^T X^T E(Y) \\ &= u^T X^T X \beta \\ &= u^T X \mu \\ &= u^T X \beta \\ &= a^T \beta. \end{aligned}$$

Thus  $u^T \hat{\theta}$  is a linear unbiased estimator of  $a^T \beta$ .

We now show that it is the BLUE.

Let  $L^T Y$  be any linear unbiased estimator of  $a^T \beta$ , where  $L \in \mathbb{R}^n$ . Then

$$E(L^T Y) = L^T X \beta = a^T \beta \quad \text{for all } \beta.$$

Since

$$a^T = u^T X,$$

we have

$$L^T X = u^T X.$$

Let

$$Q = L - Xu.$$

Then

$$Q^T X = (L - Xu)^T X = L^T X - u^T X^T X = 0.$$

Now,

$$\text{Var}(L^T Y) = \sigma^2 L^T L,$$

whereas

$$\text{Var}(u^T \hat{\theta}) = \text{Var}(u^T X^T Y) = \sigma^2 u^T X^T X u.$$

Since

$$L = Xu + Q,$$

we have

$$\begin{aligned} \text{Var}(L^T Y) &= \sigma^2 (Xu + Q)^T (Xu + Q) \\ &= \sigma^2 (u^T X^T X u + u^T X^T Q + Q^T X u + Q^T Q). \end{aligned}$$

But  $Q^T X = 0$ , and hence  $X^T Q = 0$ . Therefore,

$$\text{Var}(L^T Y) = \sigma^2 u^T X^T X u + \sigma^2 Q^T Q.$$

Thus

$$\text{Var}(L^T Y) - \text{Var}(u^T \hat{\theta}) = \sigma^2 Q^T Q \geq 0.$$

Therefore,  $u^T \hat{\theta}$  has minimum variance among all linear unbiased estimators of  $a^T \beta$ . Hence,

$$\boxed{\text{BLUE}(a^T \beta) = u^T \hat{\theta}}$$

for some  $u \in \mathbb{R}^n$ .

2. Consider the linear regression model

$$Y = X\beta + \epsilon,$$

with  $\beta \in \mathbb{R}^p$ ,  $\text{rank}(X) = p$  and

$$\epsilon \sim N_n(0, \sigma^2 I).$$

Let

$$\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_p)^T$$

be the least squares estimator of  $\beta$ .

Let  $X_{\cdot j}$  denote the  $j$ -th column of  $X$  and let  $X_{-j}$  denote the  $n \times (p-1)$  submatrix obtained by removing the  $j$ -th column of  $X$ , for  $j = 1, \dots, p$ .

(a) Show that

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{X_{\cdot j}^T (I - P_{-j}) X_{\cdot j}},$$

where  $P_{-j}$  denotes the orthogonal projector onto the column space of  $X_{-j}$ .

(b) Now consider the one-variable linear regression model (a submodel of the general model described above):

$$Y = X_{\cdot j} \beta_j + \epsilon.$$

If this submodel happens to be true, find the distribution of  $\tilde{\beta}_j$ , the least squares estimator of  $\beta_j$ .

If however the submodel is not true but the full model is true, what is the distribution of  $\tilde{\beta}_j$ ?

(c) Give an argument to say why it is preferable to use the submodel for estimating  $\beta_j$ , rather than the full model, if indeed the submodel is correct. However, if the submodel is not correct, state at least one disadvantage of using this model for statistical inference.

(d) Can you think of any circumstances when the distributions of  $\hat{\beta}_j$  from the full model and  $\tilde{\beta}_j$  from the submodel coincide?

**Solution:**

We are given the normal linear regression model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I_n),$$

where

$$\beta = (\beta_1, \dots, \beta_p)^T, \quad \text{rank}(X) = p.$$

Let  $X_j$  denote the  $j$ -th column of  $X$ , and let  $X_{-j}$  denote the  $n \times (p-1)$  matrix obtained by deleting  $X_j$ . Let  $P_{-j}$  denote the orthogonal projection matrix onto  $\mathcal{C}(X_{-j})$ :

$$P_{-j} = X_{-j}(X_{-j}^T X_{-j})^{-1} X_{-j}^T.$$

(a) The least-squares estimator of  $\beta$  is

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

Since

$$\text{Var}(Y) = \sigma^2 I_n,$$

we have

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}.$$

We partition  $X$  as

$$X = (X_{-j} \ X_j)$$

and write

$$X^T X = \begin{pmatrix} A & b \\ b^T & d \end{pmatrix},$$

where

$$A = X_{-j}^T X_{-j}, \quad b = X_{-j}^T X_j, \quad d = X_j^T X_j.$$

Since  $\text{rank}(X) = p$ , the columns of  $X$  are linearly independent. Hence  $X_{-j}$  also has full column rank, so  $A$  is nonsingular. By the block matrix inverse formula,

$$\begin{pmatrix} A & b \\ b^T & d \end{pmatrix}^{-1} = \begin{pmatrix} * & * \\ * & (d - b^T A^{-1} b)^{-1} \end{pmatrix}.$$

Therefore, the  $(j, j)$ -th diagonal element of  $(X^T X)^{-1}$  is

$$\frac{1}{d - b^T A^{-1} b}.$$

Consequently,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{X_j^T X_j - X_j^T X_{-j} (X_{-j}^T X_{-j})^{-1} X_{-j}^T X_j}.$$

Now,

$$P_{-j} = X_{-j} (X_{-j}^T X_{-j})^{-1} X_{-j}^T,$$

so

$$\begin{aligned} X_j^T (I_n - P_{-j}) X_j &= X_j^T X_j - X_j^T P_{-j} X_j \\ &= X_j^T X_j - X_j^T X_{-j} (X_{-j}^T X_{-j})^{-1} X_{-j}^T X_j. \end{aligned}$$

Thus,

$$\boxed{\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{X_j^T (I_n - P_{-j}) X_j}}.$$

Also, since  $P_{-j}$  is an orthogonal projection,

$$(I_n - P_{-j})^T = (I_n - P_{-j}), \quad (I_n - P_{-j})^2 = I_n - P_{-j}.$$

Hence

$$\begin{aligned} X_j^T (I_n - P_{-j}) X_j &= X_j^T (I_n - P_{-j})^2 X_j \\ &= \|(I_n - P_{-j}) X_j\|^2. \end{aligned}$$

Therefore, equivalently,

$$\boxed{\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(I_n - P_{-j}) X_j\|^2}}.$$

(b) Consider the one-variable submodel

$$Y = X_j \beta_j + \varepsilon.$$

Let  $\tilde{\beta}_j$  denote the least-squares estimator of  $\beta_j$  under this submodel. Then

$$\tilde{\beta}_j = (X_j^T X_j)^{-1} X_j^T Y = \frac{X_j^T Y}{X_j^T X_j}.$$

First suppose that the submodel is true. Then

$$Y = X_j \beta_j + \varepsilon,$$

and therefore

$$\begin{aligned}\tilde{\beta}_j &= (X_j^T X_j)^{-1} X_j^T (X_j \beta_j + \varepsilon) \\ &= \beta_j + (X_j^T X_j)^{-1} X_j^T \varepsilon.\end{aligned}$$

Thus

$$E(\tilde{\beta}_j) = \beta_j.$$

Moreover,

$$\begin{aligned}\text{Var}(\tilde{\beta}_j) &= \frac{1}{(X_j^T X_j)^2} X_j^T \text{Var}(\varepsilon) X_j \\ &= \frac{1}{(X_j^T X_j)^2} X_j^T (\sigma^2 I_n) X_j \\ &= \frac{\sigma^2}{X_j^T X_j}.\end{aligned}$$

Because  $\tilde{\beta}_j$  is a linear function of the normal vector  $\varepsilon$ ,

$$\tilde{\beta}_j \sim N\left(\beta_j, \frac{\sigma^2}{X_j^T X_j}\right)$$

when the submodel is true.

Now suppose instead that the full model is true:

$$Y = X_j \beta_j + X_{-j} \beta_{-j} + \varepsilon.$$

The full-model least-squares estimator satisfies

$$\hat{\beta} \sim N_p\left(\beta, \sigma^2 (X^T X)^{-1}\right).$$

Hence, by part (a),

$$\hat{\beta}_j \sim N\left(\beta_j, \frac{\sigma^2}{X_j^T (I_n - P_{-j}) X_j}\right).$$

For completeness, if the full model is true but the one-variable submodel is incorrectly fitted, then

$$\begin{aligned}\tilde{\beta}_j &= (X_j^T X_j)^{-1} X_j^T (X_j \beta_j + X_{-j} \beta_{-j} + \varepsilon) \\ &= \beta_j + (X_j^T X_j)^{-1} X_j^T X_{-j} \beta_{-j} + (X_j^T X_j)^{-1} X_j^T \varepsilon.\end{aligned}$$

Therefore,

$$E(\tilde{\beta}_j) = \beta_j + (X_j^T X_j)^{-1} X_j^T X_{-j} \beta_{-j}.$$

Thus, in general,  $\tilde{\beta}_j$  is biased for  $\beta_j$  when the submodel is false. This is the usual omitted-variable bias.

(c) Suppose now that the one-variable submodel is true. Then both estimators are unbiased for  $\beta_j$ , but their variances are

$$\text{Var}(\tilde{\beta}_j) = \frac{\sigma^2}{X_j^T X_j}$$

and

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{X_j^T (I_n - P_{-j}) X_j}.$$

Since  $P_{-j}$  is an orthogonal projection,

$$X_j^T P_{-j} X_j = \|P_{-j} X_j\|^2 \geq 0.$$

Therefore,

$$\begin{aligned}X_j^T (I_n - P_{-j}) X_j &= X_j^T X_j - X_j^T P_{-j} X_j \\ &\leq X_j^T X_j.\end{aligned}$$

Hence

$$\text{Var}(\tilde{\beta}_j) \leq \text{Var}(\hat{\beta}_j).$$

Thus, if the one-variable submodel is correct,  $\tilde{\beta}_j$  is at least as efficient as the full-model estimator  $\hat{\beta}_j$ , and is strictly more efficient whenever  $P_{-j} X_j \neq 0$ .

The reason is that

$$X_j^T (I_n - P_{-j}) X_j = \|(I_n - P_{-j}) X_j\|^2$$

is the squared length of the component of  $X_j$  orthogonal to the other regressors. If  $X_j$  has a substantial component in  $\mathcal{C}(X_{-j})$ , less independent variation is available for estimating  $\beta_j$ , increasing the variance of  $\hat{\beta}_j$ .

However, if the one-variable submodel is false, using  $\tilde{\beta}_j$  can introduce omitted-variable bias, as shown in part (b). Hence the variance advantage alone does not justify using the submodel when it is misspecified.

(d) Suppose the one-variable submodel is true. Then

$$\tilde{\beta}_j \sim N\left(\beta_j, \frac{\sigma^2}{X_j^T X_j}\right)$$

and

$$\hat{\beta}_j \sim N\left(\beta_j, \frac{\sigma^2}{X_j^T (I_n - P_{-j}) X_j}\right).$$

Since the two normal distributions have the same mean, they coincide if and only if their variances are equal. Thus,

$$X_j^T X_j = X_j^T (I_n - P_{-j}) X_j,$$

which is equivalent to

$$X_j^T P_{-j} X_j = 0.$$

But

$$X_j^T P_{-j} X_j = \|P_{-j} X_j\|^2,$$

so

$$X_j^T P_{-j} X_j = 0 \iff P_{-j} X_j = 0.$$

This holds precisely when  $X_j$  is orthogonal to  $\mathcal{C}(X_{-j})$ , i.e.

$$X_j^T X_{-j} = 0.$$

Therefore, the two estimators have the same distribution precisely when  $X_j$  is orthogonal to all the other regressors. In that case,

$$P_{-j} X_j = 0,$$

and hence

$$(I_n - P_{-j}) X_j = X_j.$$

Consequently,

$$\tilde{\beta}_j \stackrel{d}{=} \hat{\beta}_j \sim N\left(\beta_j, \frac{\sigma^2}{X_j^T X_j}\right).$$

3. Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{i,p-1} + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

(a) Show that the  $R^2$  can be expressed as

$$R^2 = \frac{\tilde{\beta}^T \tilde{X}^T \tilde{X} \tilde{\beta}}{\sum_{i=1}^n (Y_i - \bar{Y})^2},$$

where  $\tilde{X}$  is the  $n \times (p-1)$  matrix with  $(i, j)$ -th element

$$x_{ij} - x_{\cdot j}, \quad x_{\cdot j} = \frac{1}{n} \sum_{i=1}^n x_{ij},$$

and  $\tilde{\beta}$  is the BLUE for

$$(\beta_1, \dots, \beta_{p-1})^T.$$

(b) Find the distribution of

$$\frac{R^2}{1 - R^2},$$

assuming that the linear regression model above is true.

**Solution:**

We write the model as

$$Y = X\beta + \epsilon, \quad \epsilon \sim N_n(0, \sigma^2 I_n),$$

where

$$\beta = (\beta_0, \beta_1, \dots, \beta_{p-1})^T.$$

Let

$$\tilde{X} = \begin{pmatrix} x_{11} - x_{\cdot 1} & \cdots & x_{1,p-1} - x_{\cdot, p-1} \\ \vdots & & \vdots \\ x_{n1} - x_{\cdot 1} & \cdots & x_{n,p-1} - x_{\cdot, p-1} \end{pmatrix},$$

and let

$$\tilde{\beta} = (\beta_1, \dots, \beta_{p-1})^T.$$

Then the model can be reparameterised as

$$Y = \theta \mathbf{1}_n + \tilde{X} \tilde{\beta} + \epsilon,$$

where

$$\theta = \beta_0 + \sum_{j=1}^{p-1} \beta_j x_{\cdot j}.$$

Since

$$\mathbf{1}_n^T \tilde{X} = 0,$$

the column spaces of  $\mathbf{1}_n$  and  $\tilde{X}$  are orthogonal. Let

$$P_1 = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T$$

be the projection matrix onto  $\mathcal{C}(\mathbf{1}_n)$  and

$$P_X = X(X^T X)^{-1} X^T$$

be the projection matrix onto  $\mathcal{C}(X)$ . Then

$$P_X = P_1 + P_{\tilde{X}},$$

where

$$P_{\tilde{X}} = \tilde{X}(\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T.$$

(a) We have

$$\hat{Y} = P_X Y.$$

Therefore,

$$\hat{Y} - \bar{Y} \mathbf{1}_n = (P_X - P_1)Y = P_{\tilde{X}} Y.$$

The least squares estimator of  $\tilde{\beta}$  in the reparameterised model is

$$\hat{\tilde{\beta}} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T Y.$$

Hence

$$P_{\tilde{X}} Y = \tilde{X} \hat{\tilde{\beta}}.$$

Thus,

$$\begin{aligned} SSR &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \\ &= \|\hat{Y} - \bar{Y} \mathbf{1}_n\|^2 \\ &= \|P_{\tilde{X}} Y\|^2 \\ &= Y^T P_{\tilde{X}} Y \\ &= Y^T \tilde{X} (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T Y \\ &= \hat{\tilde{\beta}}^T \tilde{X}^T \tilde{X} \hat{\tilde{\beta}}. \end{aligned}$$

Also,

$$SSTO = \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

Therefore,

$$R^2 = \frac{\hat{\tilde{\beta}}^T \tilde{X}^T \tilde{X} \hat{\tilde{\beta}}}{\sum_{i=1}^n (Y_i - \bar{Y})^2}.$$

(b) Since

$$R^2 = \frac{SSR}{SSTO},$$

we have

$$\frac{R^2}{1 - R^2} = \frac{SSR}{SSE}.$$

Now,

$$SSE = Y^T (I_n - P_X) Y$$

and

$$SSR = Y^T (P_X - P_1) Y = Y^T P_{\tilde{X}} Y.$$

Since

$$E(Y) = X\beta,$$

and  $P_{\tilde{X}}$  is the projection onto the column space of  $\tilde{X}$ , we have

$$P_{\tilde{X}}E(Y) = P_{\tilde{X}}\tilde{X}\tilde{\beta} = \tilde{X}\tilde{\beta}.$$

Consequently,

$$\frac{SSR}{\sigma^2} = \frac{Y^T P_{\tilde{X}} Y}{\sigma^2}$$

has a noncentral chi-square distribution with  $p - 1$  degrees of freedom. Its noncentrality parameter is

$$\begin{aligned}\lambda &= \frac{E(Y)^T P_{\tilde{X}} E(Y)}{\sigma^2} \\ &= \frac{\tilde{\beta}^T \tilde{X}^T \tilde{X} \tilde{\beta}}{\sigma^2}.\end{aligned}$$

Thus, using the convention for the noncentral chi-square distribution adopted in the question,

$$\boxed{\frac{SSR}{\sigma^2} \sim \chi_{p-1}^2 \left( \frac{\lambda}{2} \right)}.$$

Next,

$$\frac{SSE}{\sigma^2} = \frac{Y^T (I_n - P_X) Y}{\sigma^2}.$$

Since

$$(I_n - P_X)E(Y) = 0,$$

this is a central chi-square random variable with  $n - p$  degrees of freedom:

$$\boxed{\frac{SSE}{\sigma^2} \sim \chi_{n-p}^2}.$$

We also need to establish independence of  $SSR$  and  $SSE$ . This can be shown directly, without using the Fisher-Cochran theorem.

Let

$$Q = P_X - P_1 = P_{\tilde{X}}.$$

Then  $Q$  and  $I_n - P_X$  are symmetric projection matrices, and

$$\begin{aligned}Q(I_n - P_X) &= (P_X - P_1)(I_n - P_X) \\ &= P_X - P_X^2 - P_1 + P_1 P_X \\ &= P_X - P_X - P_1 + P_1 \\ &= 0,\end{aligned}$$

where we have used

$$P_X^2 = P_X, \quad P_1 P_X = P_1.$$

Now  $Y \sim N_n(X\beta, \sigma^2 I_n)$ , so

$$QY \quad \text{and} \quad (I_n - P_X)Y$$

are jointly normally distributed. Their covariance matrix is

$$\begin{aligned}\text{Cov}(QY, (I_n - P_X)Y) &= Q \text{Var}(Y)(I_n - P_X) \\ &= \sigma^2 Q(I_n - P_X) \\ &= 0.\end{aligned}$$

Therefore,

$$QY \perp\!\!\!\perp (I_n - P_X)Y.$$

Since

$$SSR = Y^T QY = \|QY\|^2$$

and

$$SSE = Y^T (I_n - P_X)Y = \|(I_n - P_X)Y\|^2,$$

it follows that

$$\boxed{SSR \perp\!\!\!\perp SSE}.$$

Consequently,

$$\frac{SSR/(p-1)}{SSE/(n-p)} \sim F_{p-1, n-p} \left( \frac{\lambda}{2} \right).$$

But

$$\frac{SSR}{SSE} = \frac{R^2}{1 - R^2}.$$



Hence,

$$\frac{n-p}{p-1} \frac{R^2}{1-R^2} \sim F_{p-1, n-p} \left( \frac{\lambda}{2} \right),$$

where

$$\lambda = \frac{\tilde{\beta}^T \tilde{X}^T \tilde{X} \tilde{\beta}}{\sigma^2}.$$

4. Let  $I_n$  denote the identity matrix in  $\mathbb{R}^n$ . Let  $A$  be an  $n \times n$  positive semidefinite matrix such that

$$\text{rank}(A) = r$$

and

$$\text{rank}(I_n - A) = n - r$$

for some  $1 \leq r < n$ .

Prove that  $A$  is an orthogonal projection matrix.

In addition, if

$$Y \sim N(\mu, I_n),$$

then show, without using Fisher-Cochran theorem, that if

$$U_1 := Y^T A Y$$

and

$$U_2 := Y^T Y - Y^T A Y,$$

then  $U_1$  and  $U_2$  are independent, with

$$U_1 \sim \chi_r^2(\delta_1/2)$$

and

$$U_2 \sim \chi_{n-r}^2(\delta_2/2),$$

where

$$\delta_1 = \mu^T A \mu, \quad \delta_2 = \mu^T \mu - \mu^T A \mu.$$

**Solution:**

First, since  $A$  is positive semidefinite with  $\text{rank}(A) = r$ , while

$$\text{rank}(I_n - A) = n - r,$$

we have

$$\text{rank}(A) + \text{rank}(I_n - A) = n.$$

Since  $A$  is positive semidefinite, it has a spectral decomposition of the form

$$A = L_1 L_1^T,$$

where  $L_1$  is an  $n \times r$  matrix with orthonormal columns. Similarly, since  $I_n - A$  is also positive semidefinite and has rank  $n - r$ , we can write

$$I_n - A = L_2 L_2^T,$$

where  $L_2$  is an  $n \times (n - r)$  matrix with orthonormal columns.

Therefore,

$$L_1 L_1^T + L_2 L_2^T = I_n.$$

Hence

$$\begin{pmatrix} L_1 & L_2 \end{pmatrix} \begin{pmatrix} L_1^T \\ L_2^T \end{pmatrix} = I_n.$$

Let

$$P = \begin{pmatrix} L_1 & L_2 \end{pmatrix}.$$

Then  $P$  is an orthogonal matrix. Thus

$$P^T P = I_n,$$

which gives

$$\begin{pmatrix} L_1^T L_1 & L_1^T L_2 \\ L_2^T L_1 & L_2^T L_2 \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ 0 & I_{n-r} \end{pmatrix}.$$

Consequently,

$$L_1^T L_2 = L_2^T L_1 = 0.$$

Therefore,

$$\begin{aligned} A(I_n - A) &= L_1 L_1^T L_2 L_2^T \\ &= 0. \end{aligned}$$

Hence

$$A - A^2 = 0,$$

and therefore

$$\boxed{A^2 = A}.$$

Thus  $A$  is idempotent. Since  $A$  is also symmetric (as it is positive semidefinite), it follows that  $A$  is an orthogonal projection matrix. Equivalently,

$$(I_n - A)^2 = I_n - A.$$

Thus there exists an orthogonal matrix  $P$  such that

$$\boxed{A = P \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} P^T.}$$

Now let

$$Z = P^T Y.$$

Since

$$Y \sim N(\mu, I_n)$$

and  $P$  is orthogonal,

$$Z = P^T Y \sim N(P^T \mu, I_n).$$

Hence the components  $Z_1, \dots, Z_n$  are independent and

$$Z_i \sim N((P^T \mu)_i, 1).$$

Using the spectral representation of  $A$ ,

$$\begin{aligned} Y^T A Y &= Y^T P \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} P^T Y \\ &= Z^T \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} Z \\ &= \sum_{i=1}^r Z_i^2. \end{aligned}$$

For  $i = 1, \dots, r$ ,

$$Z_i \sim N((P^T \mu)_i, 1).$$

Therefore,

$$\sum_{i=1}^r Z_i^2 \sim \chi_r^2(\delta_1/2),$$

where

$$\begin{aligned} \delta_1 &= \sum_{i=1}^r (P^T \mu)_i^2 \\ &= \mu^T P \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} P^T \mu \\ &= \mu^T A \mu. \end{aligned}$$

Hence

$$\boxed{U_1 = Y^T A Y \sim \chi_r^2(\delta_1/2).}$$

Next,

$$\begin{aligned} Y^T (I_n - A) Y &= Y^T Y - Y^T A Y \\ &= \sum_{i=1}^n Z_i^2 - \sum_{i=1}^r Z_i^2 \\ &= \sum_{i=r+1}^n Z_i^2. \end{aligned}$$

For  $i = r+1, \dots, n$ ,

$$Z_i \sim N((P^T \mu)_i, 1).$$

Therefore,

$$\sum_{i=r+1}^n Z_i^2 \sim \chi_{n-r}^2(\delta_2/2),$$

where

$$\begin{aligned} \delta_2 &= \sum_{i=r+1}^n (P^T \mu)_i^2 \\ &= \mu^T P \begin{pmatrix} 0 & 0 \\ 0 & I_{n-r} \end{pmatrix} P^T \mu \\ &= \mu^T (I_n - A) \mu \\ &= \mu^T \mu - \mu^T A \mu. \end{aligned}$$

Thus

$$U_2 = Y^T(I_n - A)Y \sim \chi_{n-r}^2(\delta_2/2).$$

Finally,

$$U_1 = \sum_{i=1}^r Z_i^2$$

is a function only of

$$(Z_1, \dots, Z_r),$$

while

$$U_2 = \sum_{i=r+1}^n Z_i^2$$

is a function only of

$$(Z_{r+1}, \dots, Z_n).$$

Since  $Z_1, \dots, Z_n$  are mutually independent,

$$(Z_1, \dots, Z_r) \perp\!\!\!\perp (Z_{r+1}, \dots, Z_n).$$

Therefore,

$$U_1 \perp\!\!\!\perp U_2.$$

Hence,

$$U_1 \sim \chi_r^2(\delta_1/2), \quad U_2 \sim \chi_{n-r}^2(\delta_2/2), \quad U_1 \perp\!\!\!\perp U_2,$$

where

$$\delta_1 = \mu^T A \mu, \quad \delta_2 = \mu^T \mu - \mu^T A \mu.$$

5. Consider the model

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \quad i = 1, \dots, 8,$$

where  $\epsilon_i$  are uncorrelated with zero mean and variance  $\sigma^2$ .

Use the data given below to estimate

$$\beta_1, \quad \beta_2, \quad \sigma^2.$$

| $i$      | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
|----------|----|----|----|----|----|----|----|----|
| $Y_i$    | 82 | 79 | 74 | 83 | 80 | 81 | 84 | 81 |
| $x_{i1}$ | 10 | 9  | 9  | 11 | 11 | 10 | 10 | 12 |
| $x_{i2}$ | 15 | 14 | 13 | 15 | 14 | 14 | 16 | 13 |

**Solution:**

We are given the model

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \quad i = 1, 2, \dots, 8,$$

where

$$\epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

The design matrix is

$$X = \begin{pmatrix} 10 & 15 \\ 9 & 14 \\ 9 & 13 \\ 11 & 15 \\ 11 & 14 \\ 10 & 14 \\ 10 & 16 \\ 12 & 13 \end{pmatrix}, \quad \text{rank}(X) = 2.$$

Thus,

$$Y = X\beta + \epsilon, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}.$$

We obtain the least squares estimator by minimizing

$$\|Y - Xb\|_2^2$$

with respect to  $b$ . The normal equations are

$$X^T X b = X^T Y.$$

Since  $\text{rank}(X) = 2$ ,  $X^T X$  is invertible. Hence

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

Using the given data,

$$X^T X = \begin{pmatrix} 848 & 1168 \\ 1168 & 1632 \end{pmatrix}, \quad X^T Y = \begin{pmatrix} 6612 \\ 9194 \end{pmatrix}.$$

Therefore,

$$\begin{aligned}\hat{\beta} &= \begin{pmatrix} 848 & 1168 \\ 1168 & 1632 \end{pmatrix}^{-1} \begin{pmatrix} 6612 \\ 9194 \end{pmatrix} \\ &= \frac{1}{14712} \begin{pmatrix} 1632 & -1168 \\ -1168 & 848 \end{pmatrix} \begin{pmatrix} 6612 \\ 9194 \end{pmatrix} \\ &= \begin{pmatrix} 2.647727 \\ 3.738636 \end{pmatrix}.\end{aligned}$$

Thus,

$$\boxed{\hat{\beta}_1 = 2.6477, \quad \hat{\beta}_2 = 3.7386}.$$

Next, the fitted values are

$$\hat{Y} = X\hat{\beta}.$$

Hence

$$\hat{Y} = \begin{pmatrix} 82.5568 \\ 76.1705 \\ 72.4318 \\ 85.2045 \\ 81.4659 \\ 78.8182 \\ 86.2955 \\ 80.3750 \end{pmatrix}.$$

The residual vector is therefore

$$e = Y - \hat{Y} = \begin{pmatrix} -0.5568 \\ 2.8295 \\ 1.5682 \\ -2.2045 \\ -1.4659 \\ 2.1818 \\ -2.2955 \\ 0.6250 \end{pmatrix}.$$

Consequently, the residual sum of squares is

$$\begin{aligned}SSE &= e^T e \\ &= \sum_{i=1}^8 (Y_i - \hat{Y}_i)^2 \\ &= 28.2045.\end{aligned}$$

Since  $n = 8$  and the model contains  $p = 2$  regression parameters, the residual degrees of freedom are

$$n - p = 8 - 2 = 6.$$

Therefore, the unbiased estimator of  $\sigma^2$  is

$$\hat{\sigma}^2 = \frac{SSE}{n - p} = \frac{28.2045}{6} = 4.7008.$$

Thus, the final estimates are

$$\boxed{\hat{\beta}_1 = 2.648, \quad \hat{\beta}_2 = 3.739, \quad \hat{\sigma}^2 = 4.701.}$$

6. Consider the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 + \beta_{12} x_{i1} x_{i2} + \epsilon_i, \quad i = 1, \dots, 7.$$

The predictor variables take the following values:

| $i$      | 1 | 2  | 3  | 4  | 5 | 6 | 7 |
|----------|---|----|----|----|---|---|---|
| $x_{i1}$ | 1 | 1  | -1 | -1 | 0 | 0 | 0 |
| $x_{i2}$ | 1 | -1 | 1  | -1 | 0 | 0 | 0 |

Show that

$$\beta_0, \quad \beta_1, \quad \beta_2, \quad \beta_{11} + \beta_{22}, \quad \beta_{12}$$

are estimable and find nonmatrix algebraic forms for the estimates of these parametric functions.

Find the MSE and the standard errors of the estimates.

**Solution:**

We are given the model

$$Y = X\beta + \epsilon, \quad \epsilon \sim N_7(0, \sigma^2 I_7),$$

where

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_{11} \\ \beta_{22} \\ \beta_{12} \end{pmatrix},$$

and

$$X = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Notice that the columns corresponding to  $\beta_{11}$  and  $\beta_{22}$  are identical. Hence,

$$\text{rank}(X) = 5 < 6,$$

so  $\beta_{11}$  and  $\beta_{22}$  cannot be estimated separately. However, their sum is estimable.

We first identify the estimable parameters and then obtain their BLUEs.

Define

$$\gamma = \beta_{11} + \beta_{22}.$$

Then the model can be written as

$$Y = \tilde{X}\tilde{\beta} + \epsilon,$$

where

$$\tilde{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \gamma \\ \beta_{12} \end{pmatrix}$$

and

$$\tilde{X} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The five columns of  $\tilde{X}$  are linearly independent, so

$$\text{rank}(\tilde{X}) = 5.$$

Thus all five components of  $\tilde{\beta}$  are estimable, while  $\beta_{11}$  and  $\beta_{22}$  individually are not.

In the design matrix, the columns corresponding to  $\beta_1$ ,  $\beta_2$  and  $\beta_{12}$  are mutually orthogonal and are orthogonal to all the other columns. Moreover, each of these columns has squared norm 4. Therefore their least-squares estimators are

$$\hat{\beta}_1 = \frac{x_2^T Y}{\|x_2\|^2} = \frac{Y_1 + Y_2 - Y_3 - Y_4}{4},$$

$$\hat{\beta}_2 = \frac{x_3^T Y}{\|x_3\|^2} = \frac{Y_1 - Y_2 + Y_3 - Y_4}{4},$$

and

$$\hat{\beta}_{12} = \frac{x_6^T Y}{\|x_6\|^2} = \frac{Y_1 - Y_2 - Y_3 + Y_4}{4}.$$

Thus,

$$\boxed{\hat{\beta}_1 = \frac{Y_1 + Y_2 - Y_3 - Y_4}{4}}, \quad \boxed{\hat{\beta}_2 = \frac{Y_1 - Y_2 + Y_3 - Y_4}{4}},$$

and

$$\boxed{\hat{\beta}_{12} = \frac{Y_1 - Y_2 - Y_3 + Y_4}{4}}.$$

For the remaining estimable parameters, the normal equations are

$$X^T X \hat{\beta} = X^T Y.$$

Equivalently, using the reparameterised model,

$$\tilde{X}^T \tilde{X} \hat{\beta} = \tilde{X}^T Y.$$

The columns corresponding to  $\beta_0$  and  $\gamma = \beta_{11} + \beta_{22}$  are

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

respectively. Hence the relevant normal equations are

$$7\hat{\beta}_0 + 4\hat{\gamma} = \sum_{i=1}^7 Y_i$$

and

$$4\hat{\beta}_0 + 4\hat{\gamma} = \sum_{i=1}^4 Y_i.$$

Writing

$$\hat{\gamma} = \hat{\beta}_{11} + \hat{\beta}_{22},$$

solving these equations gives

$$\hat{\beta}_0 = \frac{Y_5 + Y_6 + Y_7}{3}.$$

Also,

$$\hat{\beta}_{11} + \hat{\beta}_{22} = \frac{Y_1 + Y_2 + Y_3 + Y_4}{4} - \frac{Y_5 + Y_6 + Y_7}{3}.$$

Therefore, the BLUEs of the estimable parameters are

$$\begin{aligned} \text{BLUE}(\beta_0) &= \frac{Y_5 + Y_6 + Y_7}{3}, \\ \text{BLUE}(\beta_1) &= \frac{Y_1 + Y_2 - Y_3 - Y_4}{4}, \\ \text{BLUE}(\beta_2) &= \frac{Y_1 - Y_2 + Y_3 - Y_4}{4}, \\ \text{BLUE}(\beta_{11} + \beta_{22}) &= \frac{Y_1 + Y_2 + Y_3 + Y_4}{4} - \frac{Y_5 + Y_6 + Y_7}{3}, \\ \text{BLUE}(\beta_{12}) &= \frac{Y_1 - Y_2 - Y_3 + Y_4}{4}. \end{aligned}$$

Notice that there is no unique BLUE of  $\beta_{11}$  or  $\beta_{22}$  individually, since neither parameter is estimable. Only their sum  $\beta_{11} + \beta_{22}$  is estimable.

For the three orthogonal estimators,

$$\text{Var}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_2) = \text{Var}(\hat{\beta}_{12}) = \frac{\sigma^2}{4}.$$

Hence

$$\text{SE}(\hat{\beta}_1) = \text{SE}(\hat{\beta}_2) = \text{SE}(\hat{\beta}_{12}) = \frac{\sigma}{2}.$$

For  $\hat{\beta}_0$ ,

$$\hat{\beta}_0 = \frac{Y_5 + Y_6 + Y_7}{3},$$

so, using the independence and equal variance of the errors,

$$\text{Var}(\hat{\beta}_0) = \frac{1}{9} \text{Var}(Y_5 + Y_6 + Y_7) = \frac{\sigma^2}{3}.$$

Therefore,

$$\text{SE}(\hat{\beta}_0) = \frac{\sigma}{\sqrt{3}}.$$

Let

$$\hat{\gamma} = \hat{\beta}_{11} + \hat{\beta}_{22} = \frac{Y_1 + Y_2 + Y_3 + Y_4}{4} - \frac{Y_5 + Y_6 + Y_7}{3}.$$

Then

$$\begin{aligned}\text{Var}(\widehat{\gamma}) &= \frac{\sigma^2}{16}(1+1+1+1) + \frac{\sigma^2}{9}(1+1+1) \\ &= \frac{\sigma^2}{4} + \frac{\sigma^2}{3} \\ &= \frac{7\sigma^2}{12}.\end{aligned}$$

Hence

$$\text{SE}(\widehat{\beta}_{11} + \widehat{\beta}_{22}) = \sigma\sqrt{\frac{7}{12}}.$$

Finally,

$$\begin{aligned}\text{Cov}\left(\widehat{\beta}_0, \widehat{\beta}_{11} + \widehat{\beta}_{22}\right) &= \text{Cov}\left(\frac{Y_5 + Y_6 + Y_7}{3}, \frac{Y_1 + Y_2 + Y_3 + Y_4}{4} - \frac{Y_5 + Y_6 + Y_7}{3}\right) \\ &= -\frac{\sigma^2}{9} \text{Var}(Y_5 + Y_6 + Y_7) \\ &= -\frac{\sigma^2}{3}.\end{aligned}$$

Thus,

$$\text{Cov}\left(\widehat{\beta}_0, \widehat{\beta}_{11} + \widehat{\beta}_{22}\right) = -\frac{\sigma^2}{3}.$$

Therefore, the corresponding standard errors are

$$\begin{aligned}\text{SE}(\widehat{\beta}_0) &= \frac{\sigma}{\sqrt{3}}, \\ \text{SE}(\widehat{\beta}_1) &= \frac{\sigma}{2}, \\ \text{SE}(\widehat{\beta}_2) &= \frac{\sigma}{2}, \\ \text{SE}(\widehat{\beta}_{12}) &= \frac{\sigma}{2}, \\ \text{SE}(\widehat{\beta}_{11} + \widehat{\beta}_{22}) &= \sigma\sqrt{\frac{7}{12}}.\end{aligned}$$